

CLINICAL DECISION SUPPORT & MULTI-OMICS DOSSIER

Patient Submitter ID: TCGA-V1-A8WT | Tumor: Prostate Adenocarcinoma (PRAD) | Stage: Stage II

TCGA Project:	TCGA-PRAD	cBioPortal Study:	prad_tcg_pan_can_atlas_2018
Gender / Age:	male / 68 yrs	Vital Status / OS:	Alive (72.8 mo)
Hybrid ML Risk:	52.2% (HIGH RISK)	VQC / Classical:	49.5% / 54.0%

FOUNDATIONAL 6-DATABASE MULTI-OMICS MUTATION PROFILE (GDC, ClinVar, COSMIC, cBioPortal, gnomAD, Ensembl)

Gene	Locus (GRCh38)	Transcript	Protein Change	ClinVar Status	COSMIC ID	gnomAD Pop AF	VAF %
AR	chrX:Xq12	ENST00000374690	p.T878A Gain of Function	Likely Pathogenic (Enzalutamide Resistance) VCV000010452	COSV57342981 FATHMM: 0.97 Pathogenic	0.000005	35.2%
PTEN	chr10:10q23.31	ENST00000371953	p.R130G / Loss Phosphatase Inactivation	Pathogenic (PI3K Disinhibition) VCV000002164	COSV51034981 FATHMM: 0.99 Pathogenic	0.000014	39.5%

QUANTUM VARIATIONAL ROTATION STATE ANGLES (θ_VQC / R_Y / CNOT)

Gene Symbol	Canonical Ensembl ID	VAF Hotspot Frequency	Variational Angle θ (Radians)	Qubit Hilbert Space Projection
AR	ENSG00000169083	35.2%	θ = 1.7100 rad (98.0°)	$ \psi\rangle = \cos(0.855) 0\rangle + \sin(0.855) 1\rangle$
PTEN	ENSG00000171862	39.5%	θ = 1.8900 rad (108.3°)	$ \psi\rangle = \cos(0.945) 0\rangle + \sin(0.945) 1\rangle$

PRECISION ONCOLOGY CLINICAL ACTION & PROTOCOL RECOMMENDATION

Recommended Therapeutic Pathway:
Multiparametric prostate MRI (PI-RADS v2), PSA doubling time tracking, and PTEN deletion surveillance.

Genomic Biomarker Summary: Identified somatic hotspot mutations in AR (p.T878A), PTEN (p.R130G / Loss). Confirmed pathological classification across ClinVar and COSMIC registries with extreme somatic rarity in gnomAD exomes.

NOTICE & RESEARCH DISCLAIMER: This report is generated by the Blazeфинix Quantum-Classical Multi-Omics Research Platform using open-access academic cancer genomics datasets (NCI GDC TCGA, ClinVar, COSMIC, cBioPortal, gnomAD, Ensembl). It is calibrated strictly for computational research, translational bioinformatics, and clinical decision support benchmarking, not for standalone autonomous medical diagnosis.